

Advertising Performance Analytics

SQL, Excel, and Power BI Analysis

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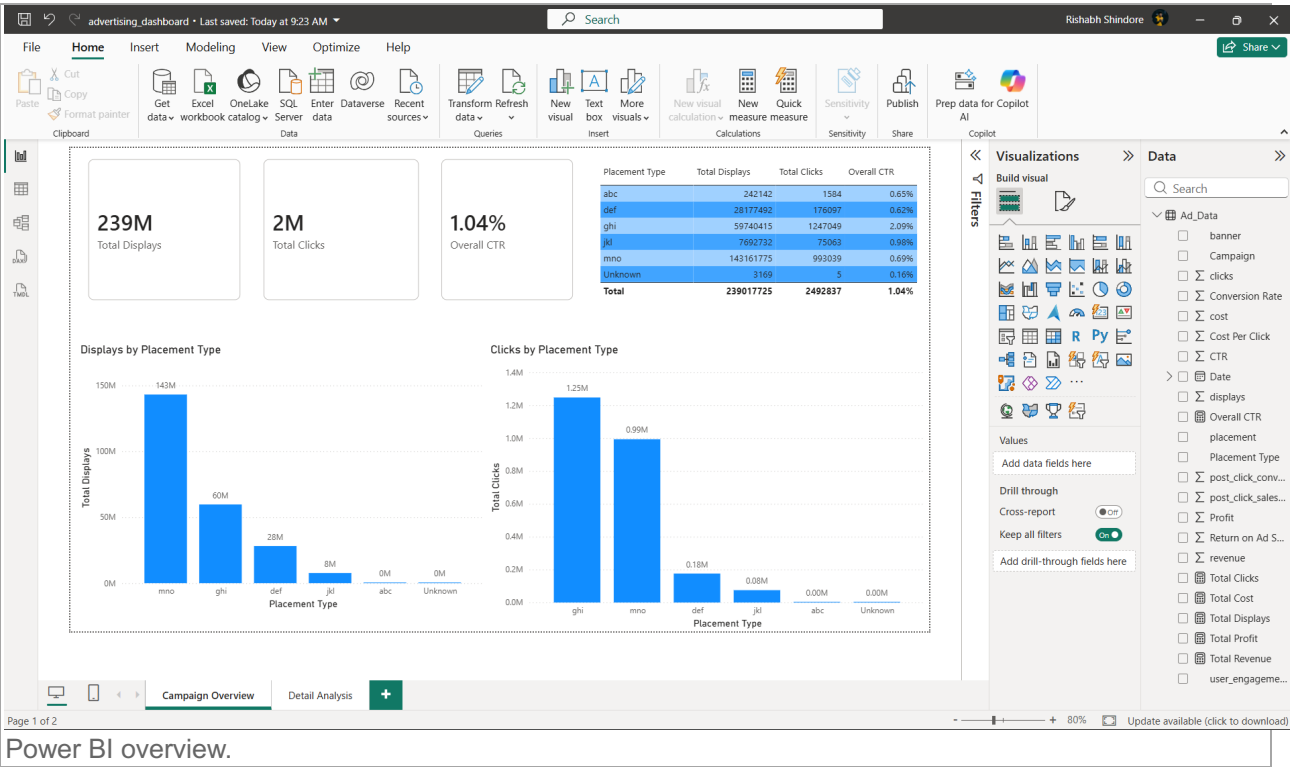
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1. Executive Summary

This project analyzes advertising performance across campaigns and placement types using SQL, Excel, and Power BI. The analysis consolidates displays, clicks, cost, revenue, profit, placement, campaign, date, banner, and user-engagement information into a validated reporting workflow.

The final Power BI report provides KPI cards, placement-level charts, a summary table, and interactive filters for campaign, date, placement type, banner, and user engagement.



2. Project Objective

The objective was to create a reliable and interactive advertising-performance analysis that answers how campaigns and placements performed, how much exposure and engagement they generated, and whether the final dashboard reconciles with the SQL and Excel results.

3. Dataset Overview

The dataset contains 15,408 advertising records. Key measures include displays, clicks, cost, revenue, and profit. Descriptive fields include date, campaign, banner, placement, cleaned placement type, and user engagement.

The analysis uses the raw data as the primary source. Summary and PivotTable outputs were used for comparison and validation rather than as the main Power BI source.

online_advertising_performance_data

Share

HomeInsertDrawPage LayoutFormulasDataReviewView

PT Placement Type Distribution

Placement	No. of Ads	% of Total Placements
abc	4501	29.21%
def	3538	22.89%
ghi	3484	22.61%
jkl	2044	18.25%
mno	988	6.38%
Unknown	413	2.68%
GrandTotal	15488	100.00%

PT Placement Type Performance Data

Placement	% of Total Clicks	% of Total Placements
abc	30.84%	29.21%
def	50.03%	22.89%
ghi	7.06%	22.61%
jkl	3.01%	18.25%
abc	0.06%	6.38%
Unknown	0.00%	0.00%
GrandTotal	100.00%	100.00%

PT Placement Type Clicks

Placement	Clicks	% of Total Clicks
abc	1247069	50.03%
def	300339	12.08%
ghi	176907	7.06%
jkl	79004	3.01%
abc	1584	0.06%
Unknown	5	0.00%
GrandTotal	2489387	100.00%

PT Campaign Performance

Campaign	Clicks	Displays	No. of Ads
camp1	14311775	14311319	6873
camp2	881158	45482269	1014
camp3	202143	3021437	8817
GrandTotal	2489387	239017725	15488

PT Campaign, Monthly Performance

Month	Clicks	Displays	CTR	Total Clicks	Total Displays	Total CTR
Apr	69895	6166272	1.14%	69895	6166272	1.14%
May	47132	5058149	0.94%	47132	5058149	0.94%
Jun	24419	3108197	0.79%	24419	3108197	0.79%
GrandTotal	1405136	14311319	0.98%	1405136	14311319	0.98%

PT Placement, Type, Monthly Performance

Month	Placement	Clicks	Displays	CTR	Total Clicks	Total Displays	Total CTR
Apr	abc	12	206	0.58%	12	206	0.58%
May	abc	25	1256	0.48%	25	1256	0.48%
Jun	abc	1543	29470	0.06%	1543	29470	0.06%
GrandTotal	1584	302143	0.06%	1584	302143	0.06%	

In April, ghi had the highest placement CTR at 2.61%, while ghi generated the most clicks with 27654. In May, ghi was the strongest placement by CTR at 1.62%. In June, ghi achieved the highest CTR at 1.36%. The results suggest that ghi was the most efficient placement overall, while ghi provided the greatest scale.

PT Placement Type Summary

Placement Type	Displays	Number of Ads	Clicks	Average Displays per Placement	CTR
abc	14311775	4501	1247069	3199.49	0.86%
def	10746415	3484	1747409	3084.24	2.09%
ghi	10746415	3538	176907	3037.44	0.62%
jkl	7900792	2044	79004	3865.18	0.98%
Unknown	14311775	988	1584	14513.57	0.08%
Total	239017725	15488	2489387	15513.57	1.04%

Unknown had the lowest average display and CTR, with approximately 6 displays per placement record and a 0.18% CTR. These records should be interpreted cautiously because the placement information was incomplete.

def delivered the greatest scale, with 143,385,779 displays and 4,501 placement records. However, ghi was the strongest placement for engagement, producing 1,247,409 clicks and the highest CTR at 2.09%.

PT Campaign Summary

Campaign Number	Number of Ads	Displays	Clicks	Average Displays per Campaign	CTR
camp1	6873	14311319	1405136	2089.27	0.98%
camp2	1014	45482269	881158	28186.54	1.94%
camp3	8817	3021437	202143	7299.13	0.60%
TOTAL	15488	239017725	2489387	15513.57	1.04%

Campaign 1 generated the highest volume, with 143,313,319 displays and 1,405,136 clicks. Campaign 2 was the most efficient campaign, achieving the highest CTR at 1.94% and the highest average clicks per campaign at 28,186.54. Campaign 3 contained the highest number of ads, with 8,817, but all ads alone did not make it the strongest performer.

online_advertising_performance

Sheet1

ReadyAccessibility: Investigate

Excel Data Sheet Screenshot

4. Data Preparation and Quality Checks

Data was reviewed for field names, data types, blank columns, duplicate calendar fields, and missing placement values. The Date field was retained as the primary date field, while redundant day and month fields were removed where they duplicated information available from Date.

The 413 records with missing placement values were retained and labeled Unknown so their valid performance measures were not excluded from the analysis. Unused blank columns were removed where applicable.

advertising_dashboard

File Home Transform Add Column View Tools Help

Close & Apply * Source * Settings * Data Source * Parameters * Output Data * Query * Properties * Advanced Editor * Refresh * Preview * Manage * Choose * Remove * Columns * Columns * Keep * Remove * Rows * Rows * Sort * Split * Group * By * Data Type: Date * Use First Row as Headers * Replace Values * Merge Queries * Append Queries * Combine Files * Combine

Queries [1]

Ad_Data

Table.ReplaceErrorValues("Replaced Value", {"placement", "Unknown"})

Date	Campaign	user_engagement	placement	displays	cost
01-04-2020	camp 1	High	abc	20170	4
01-04-2020	camp 1	High	def	14701	17259
01-04-2020	camp 1	High	ghi	552	16
01-04-2020	camp 1	Low	def	2234	2963
01-04-2020	camp 1	Low	ghi	580	2188
01-04-2020	camp 1	Low	mno	20152	56499
01-04-2020	camp 1	Medium	def	392142	62395
01-04-2020	camp 1	High	jkl	214074	872
01-04-2020	camp 1	Low	def	1188	980
01-04-2020	camp 1	Low	jkl	1769	6195
01-04-2020	camp 1	Medium	def	25765	4595
01-04-2020	camp 1	Medium	mno	18518	11
01-04-2020	camp 1	High	abc	52381	62042
01-04-2020	camp 1	High	ghi	10001	277579
01-04-2020	camp 1	High	mno		

17 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

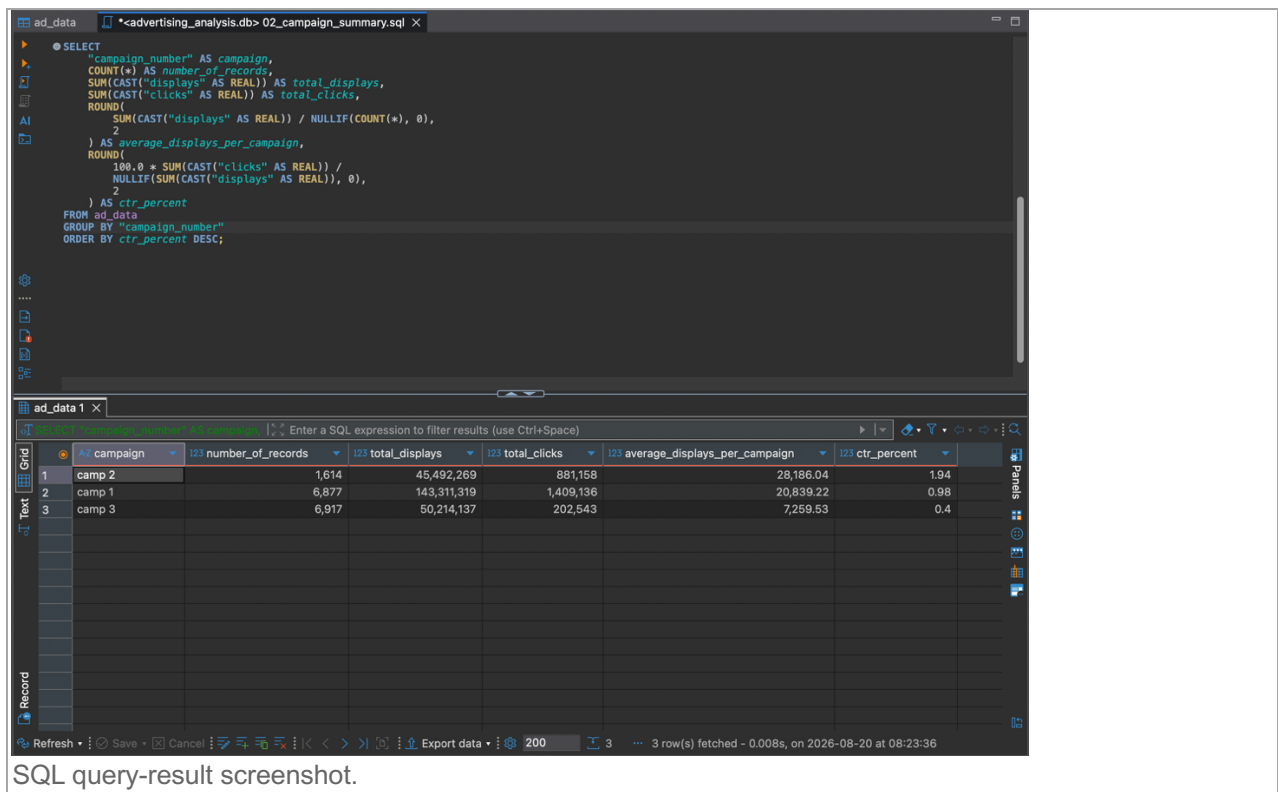
PREVIEW DOWNLOADED AT 08:42

Power Query screenshot.

5. SQL Analysis

SQL queries were used to validate the row count, inspect the source fields, identify placement categories, summarize placement and campaign performance, review monthly performance, and calculate overall totals and CTR.

The main KPI calculations use total clicks divided by total displays. The SQL scripts are included in the project SQL folder and provide a reproducible audit trail for the analysis.



6. Excel Analysis

Excel PivotTables were used as an intermediate analytical and reconciliation layer. They summarized performance by placement, campaign, and time period and provided a reference point for validating the SQL and Power BI outputs.

The Excel workbook is included in the project Data folder as supporting evidence.

8. Key Findings

The final validated totals are shown below. These figures represent the complete dataset with no slicer restrictions applied.

Advertising Performance Analytics

Project Notes and Key Findings

This section documents the dataset, data-quality treatment, placement performance, campaign performance, and validated totals for inclusion in the analytics report

Dataset and Data Quality

The dataset contains **15,408 records** across six cleaned placement categories: **abc**, **def**, **ghi**, **jkl**, **mno**, and **Unknown**.

The placement field contains **413 text values labelled #N/A**, representing unavailable placement information. These values were retained and categorised as **Unknown** for analysis so that their associated performance measures were not excluded.

Validated dataset totals

Metric	Result
Total records	15,408
Total displays	239,017,725
Total clicks	2,492,837
Overall CTR	1.04%
Unknown placement records	413

Placement Representation and Delivery

The **mno** placement had the highest representation, accounting for **4,501 records**, or **29.21%** of the dataset.

mno also received the highest number of ads served, with **143,161,775 displays**, representing **59.90%** of all displays.

Display volume was therefore highly concentrated in **mno**. Although **mno** represented only 29.21% of placement records, it accounted for 59.90% of total displays. By contrast, **def**, **jkl**, and **abc** represented larger shares of records than displays, indicating lower display volume per record.

Average displays per placement record

Placement	Total displays	Records	Average displays per record
mno	143,161,775	4,501	31,807
ghi	59,740,415	3,538	16,885

Unknown	3,169	413	8
Other categories	See SQL/Power BI output	—	Lower than mno and ghi

Interpretation

Ad delivery was strongly concentrated in mno. This placement was the primary high-volume placement, while abc and Unknown contributed very little display volume per record. Unknown should be interpreted cautiously because its placement information was unavailable.

Clicks and Engagement

The **ghi** placement generated the highest number of clicks, with **1,247,049 clicks**, representing **50.03%** of all clicks.

ghi generated 50.03% of all clicks from 24.99% of displays, while mno generated 39.84% of clicks from 59.90% of displays. This indicates stronger click concentration in ghi and higher display volume but relatively lower click contribution in mno.

Placement CTR ranking

Rank	Placement	CTR
1	ghi	2.09%
2	jkl	0.98%
3	mno	0.69%
Lowest	Unknown	0.16%

Interpretation

mno delivered the greatest scale, with 143,161,775 displays and 4,501 placement records. However, ghi was the strongest placement for engagement, producing 1,247,049 clicks and the highest CTR at 2.09%. Unknown had the lowest average delivery and CTR, with approximately 8 displays per record and a 0.16% CTR.

Campaign Performance

Campaign 1 generated the highest volume, with **143,311,319 displays** and **1,409,136 clicks**.

Campaign 2 was the most efficient campaign, achieving the highest CTR at **1.94%** and the highest average displays per campaign at **28,186.04**.

Campaign 3 contained the highest number of ads, with **6,917**; however, ad count alone did not make it the strongest performer.

Interpretation

Campaign 1 was the scale leader, while Campaign 2 was the efficiency leader. Campaign 3 had the highest record count but should not be evaluated using volume alone.

Monthly Campaign Performance

Campaign 2 was the strongest campaign in April, achieving the highest observed monthly CTR of **1.94%**. However, Campaign 2 was not active during May or June.

Campaign 1 achieved the highest CTR in both May and June, indicating that it became the leading active campaign after April.

Campaign 3 recorded the lowest observed CTR at **0.37%** in April and should be reviewed for possible optimization.

Recommended interpretation

Campaign performance changed by month. The strongest campaign overall should therefore be interpreted alongside campaign availability and the period in which each campaign was active.

Overall Conclusions

The analysis shows a clear distinction between scale and efficiency. mno was the dominant delivery placement, while ghi generated the strongest engagement response. At campaign level, Campaign 1 generated the greatest volume, whereas Campaign 2 achieved the strongest efficiency when active.

The Unknown category should remain visible in reporting because it contains valid records, but its results should be treated cautiously until placement information can be recovered or improved at source.

Recommendations

Prioritize investigation of ghi to understand the factors contributing to its 2.09% CTR and high click contribution.

Review the role of mno: it provides substantial scale, but its 0.69% CTR suggests an opportunity to improve engagement efficiency.

Audit Campaign 3, particularly its April performance, where the observed CTR was 0.37%.

Improve placement data capture to reduce the 413 Unknown records and strengthen placement-level decision-making.

Monitor both delivery volume and CTR so that high-scale placements are not evaluated on volume alone.

9. Validation Results

The SQL, Excel, and Power BI results were reconciled. The Power BI KPI cards matched the final SQL validation totals, confirming that the imported data and measures were working as intended.

The report was also checked after clearing slicers, and the Unknown placement category remained visible for the 413 records with missing placement.

10. Limitations and Recommendations

The analysis depends on the quality and completeness of the source fields. Missing placement values were handled as Unknown, which preserves the records but limits placement-specific interpretation for those rows.

Future improvements could include a dedicated date table, documented data-refresh procedures, additional campaign profitability measures, and a standardized source-data dictionary.

11. Conclusion

The project produced a complete advertising-performance workflow using SQL for structured analysis, Excel for reconciliation, and Power BI for interactive reporting. The final dashboard is supported by validated totals and documented handling of missing values.

12. Data Source & Disclaimer

This project uses a publicly available advertising dataset downloaded from Kaggle for educational and portfolio purposes. The dataset is not a live production database and should not be interpreted as representing actual business operations, real customer activity, or current advertising performance.

The data may be synthetic, anonymised, simulated, or otherwise prepared for learning and analysis. All findings in this report apply only to the supplied dataset and should not be generalized to a real organization without independent validation.

The dataset was cleaned and transformed for analysis. In particular, 413 records with unavailable placement information were categorized as Unknown. The SQL database, Excel workbook, and Power BI report were created as part of this analytical exercise.

Dataset credit and licensing information should follow the original Kaggle dataset page and its stated license. The original dataset creator and Kaggle source should be acknowledged wherever the dataset is used. Kaggle recommends checking the dataset's license and attribution requirements before reuse.

<https://www.kaggle.com/datasets/naniruddhan/online-advertising-digital-marketing-data?resource=download>

Dataset Title - Online Advertising Digital Marketing Data

Dataset Creator - N. aniruddhan

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